Collective learning

Hermes supplies memory, history, skills, and a skill ledger. This project adds owner review, export approval, and a reviewed release path.

CONTENTS

1. Private and shared state
2. Local learning and review
3. Approval and release

PRIVATE AND SHARED STATE

Learning changes the Worker's context, not model weights.

PRIVATE MEMORY

`USER.md` and `MEMORY.md` retain compact preferences and durable working facts.

PRIVATE SKILLS

`skills/private/` stores project-, customer-, or assignment-specific playbooks.

ROLE SKILLS

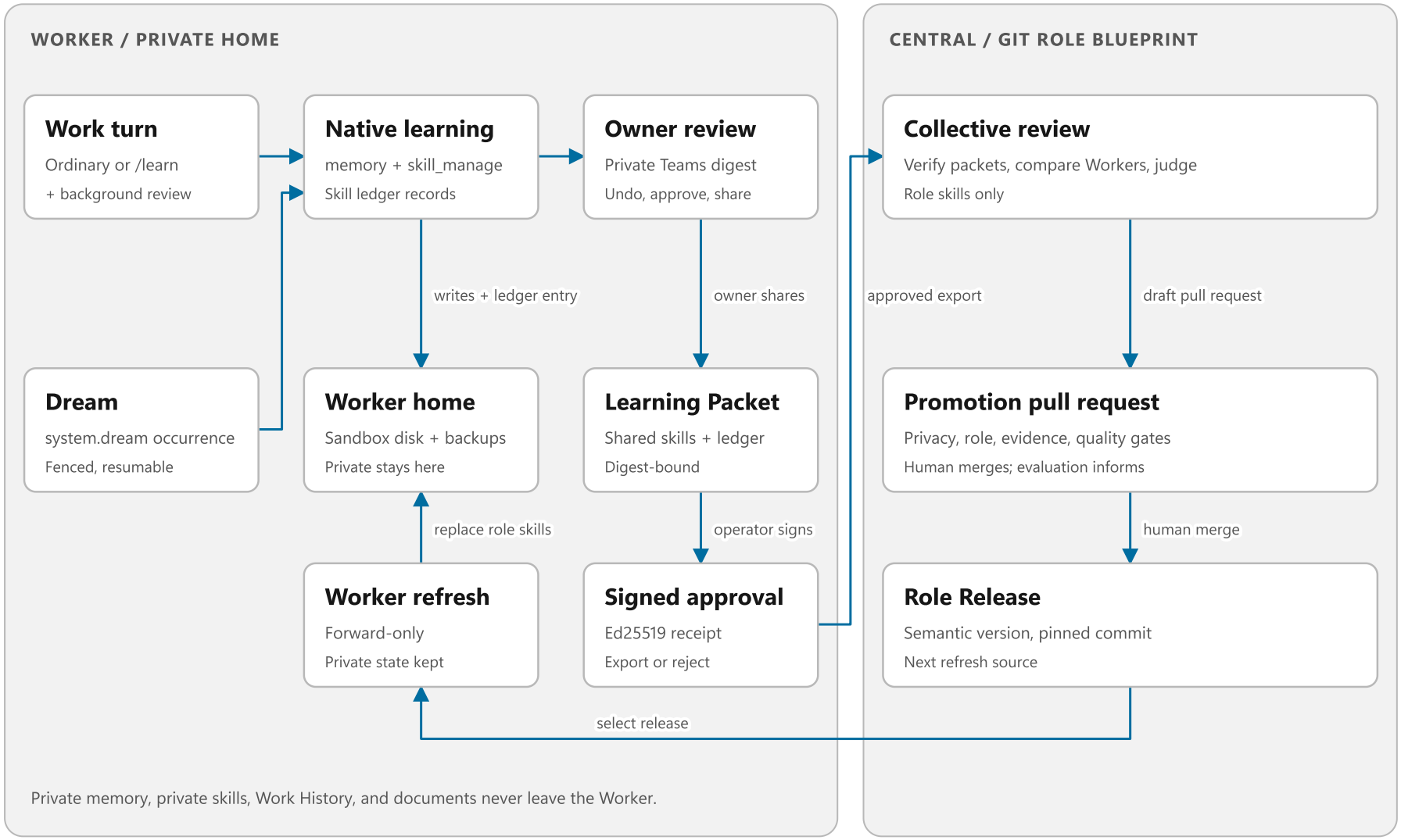
`skills/role/` starts from a reviewed Role Release and can receive governed local patches.

CANDIDATES

`skills/candidates/` holds generalized capabilities proposed locally for later review.

Raw Work History stays private. Only role and candidate changes whose current content the Worker owner chose to share enter a Learning Packet. Generalization means removing private facts, not just moving the same text into another directory.

01 / Learning lifecycle



Local learning, the owner's Share choice, and export approval are separate decisions. Nothing crosses into the central zone without a signed approval.

SOURCE

Classification policy · Owner review and private-session guard · Learning Packet 3.0

01

Load compact context; retrieve deeper history when needed.

Native Hermes supplies the mechanisms. The Role Blueprint supplies their policy.

- **Fresh session:** active `SOUL.md` and compact memory provide starting context.
- **Relevant task:** skill names and descriptions allow discovery; full instructions load when selected.
- **Recall or Dream:** session search retrieves relevant Work History instead of injecting the entire database.

Memory files and private playbooks are Worker-wide. Session keys identify the caller, but native group transcripts are shared and rotate hourly by default. A user-specific key does not create a separate memory store.

SOURCE

Native runtime setup · Home layout · Conversation boundary

LOCAL LEARNING AND REVIEW

02

A turn can learn without approving its own export.

Ordinary work, background review, and /learn write through the same native tools.

Hermes learns natively. An ordinary turn may update memory or skills, and a background review after the turn may add more; memory and skill nudges arrive every ten turns. Hermes' native `/learn` turns a request into a skill through `skill_manage`.

The native skill ledger records every skill change with its actor, action, before/after hashes, and content.

A useful local improvement can remain active in this Worker before any shared release. That local acceptance neither exports private data nor grants permission to publish the skill to other Workers.

Attachment and persistence-disabled turns run in a separate native session marked `private:`. A `pre_tool_call` hook blocks `memory`, `skill_manage`, and file writes into `skills/` or `memories/` there, including in the background review fork. Separate native session identity is part of the privacy boundary, not a second shared memory.

> SOURCE
>
> Turn routing and learning instructions · Private-session hook

03

Dream reflects across work, under the same rules.

A bounded reflection task, not a second autonomous runtime.

Review
Inspect outcomes, corrections, skills, and relevant history.

Classify
Keep private context private; generalize reusable behavior.

Write or skip
Use native memory and skill tools, or store nothing.

The reviewed Dream Role Skill guides reflection. A native Service Bus receiver wakes the suspended gateway for a system occurrence; no gateway timer or receive thread owns the wait. Durable fenced phases recover completed responses; an unknown interrupted model outcome stops rather than triggering a blind replay. Ad-hoc execution does not consume the production occurrence. After a Dream, the Worker owner receives the learning digest.

> SOURCE
>
> Dream instructions · Durable coordinator

04

The Worker owner decides what stays and what may be shared.

A private Teams digest shows native ledger diffs with typed decisions.

1 **Read the digest**

After Dreaming, and on `/learning` or `/learned` in the owner's personal chat, the gateway sends a private card with recent skill changes and their diffs. Nothing is sent to group chats.

2 **Undo or approve**

Undo uses native rollback. With `HERMES_SKILL_WRITE_APPROVAL`, skill writes are staged until the owner chooses **Approve** or **Reject**.

3 **Share, keep, or discard**

For a Role Skill or Candidate Improvement change, **Share** makes its current content eligible for a packet. **Keep** copies it to the private `<skill>-playbook` and restores the release baseline; **Discard** restores the baseline.

<details>
<summary>Inspect the decision contract</summary>

The owner is `worker_owner_id`, normally the Agent Identity sponsor. Card actions are encrypted, expiring tokens bound to the owner and conversation. Every decision is an idempotent receipt in `learning/receipts/`; share choices live in `learning/share-choices.json`. A later change to a shared skill needs a new Share choice.

Background review and Dreaming write through the same native tools, so their changes appear in the same digest. A governed change without a complete ledger trace, such as a direct file edit, is shown but stays out of packets.
</details>

SOURCE

Owner decisions and rollback · Internal learning routes · Teams digest card

05

Package only what the owner shared, with its ledger chain.

Learning Packet 3.0 carries shared skill files, their native history, and a verifiable identity.

Packet preparation includes only Role Skill and Candidate Improvement changes whose current content the owner chose to share. Each carries its single `SKILL.md`, release-baseline hashes, artifact hash, native ledger chain from the release baseline to the current files, share decision, and owner approvals of staged writes. Worker/release attribution and the governed-state hash bind it to this Worker; the digest binds the packet's exact content.

Preparation lists every other governed change under `excluded` with a reason; it stays local. Only one bounded UTF-8 `SKILL.md` whose frontmatter names its directory is transferable; scripts, references, binary payloads, credentials, memories, sessions, and private playbooks never are.

A privacy scan rejects addresses, identifiers, internal URLs, secrets, and absolute paths. Packets carry no agent-proposed test scenarios.

<details>
<summary>Inspect the ledger chain</summary>

The native skill ledger records each skill write with its actor, action, before/after hashes, and content. For a shared skill, the packet carries the entries that lead from the Role Release baseline hash to the current file hash. Preparation recomputes file hashes from disk; the ledger explains them rather than vouching for them.

A change whose chain is incomplete, or whose content changed after the owner's Share choice, is excluded instead of exported.
</details>

> SOURCE
>
> Packet 3.0 preparation · Packet contracts

06

Approval binds the packet and the current Worker state.

A later edit makes the prior approval stale.

The runtime verifies an Ed25519-signed receipt against the packet digest, Worker, release commit, and governed-state hash. A signature over a different or stale packet cannot authorize export. The gateway holds the private key; runtime and review receive public keys only.

The operator authenticates through Entra as the Worker owner. The gateway derives the approver from that verified identity; a runtime API key or caller-supplied name cannot approve a packet. Owner-bound Teams skill decisions remain a separate workflow.

APPROVE EXPORT

Allows the exact reviewed packet to leave the Worker; it does not approve merging or release.

REJECT AND REFRESH

Records the signed rejection and permits discarding those governed local changes during the controlled refresh.

An unreviewed packet must not be treated as approved simply to unblock a refresh.

> SOURCE
>
> Receipt verification and refresh guard · Operator approval flow

07

Only independent cases can test the claimed improvement.

Run baseline and candidate on the same cases and report the scope honestly.

Packets carry no agent-proposed scenarios: otherwise the author can describe a skill and the easiest test for that same description. An independently authored suite is mandatory; owner-authored holdout cases are the intended judge.

The evaluation script invokes real Hermes in isolated baseline and candidate profiles with identical non-skill state and model settings, and records case outcomes and usage. Missing usage is an error, not zero cost. Results need the case set, model/settings, baseline, candidate, and packet binding to be interpretable.

Inspect what this evaluator measures

The current runner injects selected skill text into the prompt and asserts response text with `contains`, `not_contains`, or `equals`. Tools, hooks, plugins, MCP, and background learning are disabled. It does not measure native skill discovery, live MCP workflows, tool side effects, repeated-trial variance, or general task success. Neither a signature nor a higher pass rate broadens the tested scope.

SOURCE

Evaluation runner · Evaluation contracts

08

Share through a Role Release, not by copying another Worker's home.

Preserve private continuity while replacing governed behavior.

Review packet
Check privacy, evidence, skill quality, and role alignment.

Publish release
Pin the reviewed Role Blueprint to an immutable commit.

Refresh Worker
Resolve local governed changes, then install the selected release.

A forward-only refresh preserves private memories, private playbooks, and Work History. It replaces Role Skills and archives the prior release's candidates and learning decisions. It does not copy another Worker's private home or silently merge unreviewed local edits.

SOURCE

Promotion review workflows · Refresh guard · Release installation

> **Learn natively, export only what the owner shared and a person approved, and share behavior through a reviewed Role Release.**

▼ Work turn

Ordinary or /learn

Native memory and skill tools can change the Worker home during ordinary work, and a background review after the turn may add more. Hermes' native `/learn` turns a request into a skill through `skill_manage`. Attachment and persistence-disabled turns run in a `private:` session where durable memory and skill writes are blocked.

▼ Dream

Scheduled reflection

A system schedule delivers `system.dream` through Service Bus and the gateway. The reviewed Dream Role Skill reviews recent work, classifies observations, and writes through native tools or stores nothing. Durable fenced phases recover completed responses; an unknown interrupted outcome stops for inspection. The owner receives the learning digest afterwards.

▼ Native learning

Hermes-native

Memory and `skill_manage` writes go directly to the Worker home; memory and skill nudges arrive every ten turns. The native skill ledger records every skill change with its actor, action, before/after hashes, and content. With `HERMES_SKILL_WRITE_APPROVAL`, skill writes are staged until the owner approves them.

▼ Worker home

Sandbox disk with native backups

`/data/hermes` holds `USER.md`, `MEMORY.md`, `skills/private/`, `skills/role/`, `skills/candidates/`, the skill ledger, `learning/receipts/`, sessions, and schedules. Native backups on a Blob volume carry it into a replacement runtime. Private layers never enter a Learning Packet.

▼ Owner review

Private Teams digest

Only the Worker owner receives it, after Dreaming or on `/learning`. *Undo* uses native rollback; staged writes take *Approve* or *Reject*. Role and candidate changes take *Share*, *Keep* (private playbook, baseline restored), or *Discard* (baseline restored). Actions are expiring owner-bound tokens with idempotent receipts.

▼ Learning Packet 3.0

Digest-bound export candidate

Contains only owner-shared Role Skill and Candidate Improvement changes: single `SKILL.md` files, release-baseline hashes, artifact hash, native ledger chain, share decision, and staged-write approvals. Hashes are recomputed from disk; every other change is listed as excluded with a reason. A privacy scan covers the content.

▼ Signed approval

Ed25519 receipt

The runtime verifies the signature against the packet digest, Worker, release commit, and governed-state hash. *Approve export* lets the exact packet leave; *reject and refresh* permits discarding those local changes. The gateway holds the private key.

▼ Collective review

Central workflow

Verifies packet digests and receipts, compares packets from several Workers, and judges which role skills generalize. Private skills and raw history are not inputs.

▼ Promotion pull request

Draft pull request with agentic gates

GitHub Agentic Workflows triage the change and review privacy, role alignment, evidence, and skill quality. Evaluation results inform the reviewer; a human merges.

▼ Role Release

Pinned Git commit

A semantic version pinned to an immutable commit of the Role Blueprint. It becomes the baseline for later Learning Packets.

▼ Worker refresh

Forward-only

Requires local governed changes to be exported or rejected first. Replaces role skills, archives prior candidates and learning decisions, and keeps private memory, private skills, and Work History.